When Legal Grounds Expire

Warrant Withdrawal, Dependency Cascades, and Living Normative Memory in CriptoIUS

Ignacio Adrián Lerer

Attorney (UBA), Executive MBA (IAE, Universidad Austral)

Buenos Aires, Argentina

Email: adrian@lerer.com.ar

ORCID: 0009-0007-6378-9749

Preprint, 17 August 2026

Abstract

Legal artificial intelligence is commonly evaluated at the level of the answer: whether a proposition is correct, a citation exists, or a prediction matches an outcome. This paper argues that consequential legal systems require a different object of governance: the continuing warrant for institutional reliance. A legal conclusion may have been well grounded when produced and yet become unfit for reliance after a statute is amended, a precedent is displaced, an authorization expires, a jurisdiction changes, or an evidentiary premise is withdrawn. The relevant problem is therefore not only how to store legal knowledge, but how to represent its dependencies and how to identify the conclusions affected when a ground ceases to hold.

The paper develops this argument by placing Vitaly Reznik's Zero-Trust Logic and warrant-containment experiments in dialogue with CriptoIUS, a proposed infrastructure for versioned and evolutionary legal interpretation. Reznik's work shows how withdrawal can propagate prospectively through recorded dependency graphs, while also establishing strict limits: the instrument does not discover missing dependencies, prove that external grounds are genuinely independent, detect deception, or determine whether a legal authority remains valid. Its formal outputs are machine-checked classifications, not legal interpretations. Incomplete graphs can err in either direction. Missing shared origins can make grounds appear more independent than they are, while missing alternative grounds can make a conclusion appear unsupported when it remains warranted.

Building on those results, this paper proposes a five-layer architecture for living normative memory: authority admission, dependency representation, formal containment, runtime admissibility, and institutional reliance. It also proposes append-only revalidation events for CriptoIUS, together with explicit states for uncertainty, constraint, escalation, and refusal. The central claim is limited but consequential: formal dependency containment can compute what follows from a recorded change, but only legal and institutional governance can decide whether the change occurred, what authority it carries, and whether the result may legitimately support action. A trustworthy legal memory system must preserve both functions without collapsing one into the other.

Keywords: legal AI; normative memory; dependency graphs; warrant containment; runtime admissibility; institutional reliance; CriptoIUS; provenance; legal interpretation; authority

1. Introduction

Legal systems remember. They remember statutes, judgments, administrative acts, contractual interpretations, evidentiary findings, and institutional practices. Digital legal systems remember them more efficiently, but not necessarily more responsibly. Most repositories preserve documents. More advanced systems preserve citations, entities, embeddings, and links. Yet a legal conclusion is not merely a piece of text connected to another piece of text. It is a claim whose permissible use depends on an authority, a factual record, a jurisdiction, a temporal state, an interpretive lineage, and a professional decision about reliance.

This distinction becomes critical when legal artificial intelligence moves from retrieval to recommendation, and from recommendation to institutional action. A system may accurately reproduce a proposition that was valid yesterday but no longer governs today. It may cite an authentic decision whose holding has been displaced, whose jurisdiction is irrelevant, or whose factual predicate does not survive into the present matter. It may preserve an unbroken technical record while the legal basis for reliance has quietly expired.

The ordinary response is to demand better updating, more citations, or a more complete audit trail. Each is necessary. None is sufficient. Updating tells us that a source changed. Citation tells us where a proposition came from. An audit trail tells us what the system did. The unresolved question is what should happen to every downstream conclusion that depended on a ground that is no longer entitled to support it.

That is a problem of warrant containment. It is also a problem of normative memory. A memory system worthy of institutional reliance must be able to represent not only that a proposition was stored, but why it was accepted, what it depended on, which conditions limited it, and what later events require its revalidation. The system should identify affected conclusions prospectively, before they are relied upon again, rather than merely reconstructing the error after a consequence has occurred.

Vitaly Reznik's *Warrant Containment in Dependency Graphs* offers a useful formal and experimental vocabulary for this problem. Its core discipline is that truth is not granted on credit. An unverified atom does not acquire truth merely by entering a calculation. More importantly for legal systems, the withdrawal of a recorded ground can propagate through the conclusions that depend on it. The work therefore provides machinery for answering a bounded question: given a represented dependency structure and a change supplied to the system, what else is affected? The discussion relies particularly on the containment results and stated limitations in sections 3.6 and 6 of Reznik (2026).

The boundary of that contribution is as important as the contribution itself. The instrument cannot determine whether a precedent has legally lost authority. It cannot know that two apparently distinct reports reproduce the same hidden source. It cannot discover a dependency that nobody recorded. It cannot transform technical provenance into a certificate of legal legitimacy. Those are not defects to be concealed. They locate the legal and institutional layer that must surround formal containment.

This paper develops that legal layer in dialogue with the public research program of CriptoIUS, which studies versioned and evolutionary legal interpretation. CriptoIUS is used here as a conceptual frame, not as a description of a commercial implementation. The paper does not disclose or prescribe product workflows, internal schemas, decision rules, scoring mechanisms, or deployment architecture. Nor does it claim that the proposed technical integration has been built or validated.

The resulting thesis is straightforward. Formal systems can compute the consequences of a represented withdrawal. Legal institutions must decide what counts as a withdrawal, whether the affected conclusion remains admissible, and who may rely on it. Living normative memory requires both, joined but not conflated.

2. From Document Storage to Normative Memory

2.1 A legal proposition is more than text

A conventional legal database treats a judgment, statute, or opinion as a document. Retrieval systems improve access by indexing terms, entities, citations, and semantic similarity. These functions answer questions about location and resemblance. They do not by themselves answer whether a retrieved proposition remains fit to govern the present decision.

Consider a simple proposition: a contractual notice period is thirty days. That statement may derive from a statutory provision, a contract clause, an administrative rule, a judicial interpretation, or a professional synthesis of several sources. Its legal force may depend on the identity of the parties, their jurisdiction, the date of the contract, the nature of the relationship, mandatory law, and later amendments. A system that stores only the sentence and its citation has preserved information, but not the conditions of lawful reliance.

Normative memory therefore differs from semantic memory. Semantic memory asks what a proposition means and how it relates conceptually to other propositions. Normative memory must also preserve why an institution treated the proposition as action-guiding. Its relevant metadata are not ornamental. Authority, applicability, temporal validity, jurisdiction, evidentiary basis, review status, and interpretive lineage determine whether the proposition can legitimately enter a decision.

2.2 Memory must preserve change without rewriting history

Law changes through amendment, repeal, overruling, distinction, displacement, administrative reinterpretation, contractual modification, and changes in the facts that make a norm applicable. A reliable system should not overwrite the old proposition as though it had never existed. The earlier interpretation may remain legally relevant to acts performed while it governed, to pending proceedings, to transitional rules, or to the reconstruction of institutional responsibility.

CriptoIUS addresses this through versioned, append-only interpretive events. An interpretation may be refined, distinguished, displaced, overruled, or marked as affected by temporal change or jurisdictional divergence. The prior state remains visible. The later state records what changed and under whose authority. This model is essential for retrospective explanation, but the present paper adds a prospective requirement: when a ground changes, the system should identify the interpretations and decisions whose warrants may no longer survive.

2.3 The difference between provenance and warrant

Provenance identifies origin and lineage. Warrant concerns entitlement to rely. An authentic document may lack authority. An authoritative source may not apply to the matter. An applicable rule may depend on a fact that is disputed or no longer true. A conclusion can therefore have excellent provenance and an insufficient warrant.

This distinction prevents a common category error in legal technology. Cryptographic integrity, authenticated origin, and immutable registration can prove that a record existed in a particular form at a particular time. They do not prove that its content was legally correct, that its source governed the decision, or that reliance remained justified when the output was used. CriptoIUS accordingly treats registration as evidence of a versioned assertion, not as a certificate of legal truth.

2.4 Normative memory as an extended phenotype

The evolutionary account of CriptoIUS adds a further dimension. An interpretation does not remain confined to the judgment, opinion, or contract in which it first appeared. It reproduces through later decisions, templates, professional training, compliance manuals, software defaults, and the categories through which new disputes are framed. In this limited sense, a legal interpretation has effects beyond its original institutional vehicle. It helps shape the environment in which later actors reason and decide.

Extended Phenotype Theory (EPT) is used here as an explanatory model for that cross-institutional persistence. The claim is not that legal propositions are biological genes or that legal development is reducible to natural selection. The relevant analogy concerns inherited effects. A source interpretation can alter downstream institutional structures that then reproduce the same interpretive pattern, sometimes without preserving an explicit citation to the originating source.

This creates a problem that ordinary document updating cannot solve. A precedent may be overruled while its categories remain embedded in contracts, forms, model clauses, compliance tools, and machine-generated legal outputs. The textual source has lost authority, but its phenotype continues to organize behavior. The same can occur when a legal transplant survives in a receiving system after the conditions that initially justified the transplant have changed.

Normative memory must therefore track two different forms of persistence. The first is documentary persistence: the source remains stored and retrievable. The second is phenotypic persistence: the interpretive pattern continues to shape later decisions and artifacts. Withdrawal from the first does not automatically eliminate the second. A dependency-aware system should identify where the displaced interpretation has been inherited and require those later uses to be examined under their own authority and context.

This also limits evolutionary fitness as a governance criterion. Replication can show that an interpretation is useful, familiar, institutionally advantageous, or difficult to dislodge. It does not show that the interpretation remains legally valid or legitimate. A highly replicated interpretation may persist precisely because it is embedded in templates and infrastructure that few actors revisit. Normative fitness describes propagation; runtime admissibility governs reliance.

3. What Warrant Containment Adds

3.1 Verification is attached to grounds, not borrowed by conclusions

Reznik's Zero-Trust Logic begins from a refusal to treat absence of verification as truth. Its third symbol marks an unverified atom; it is not a truth value and is barred from the value of every compound. Verdicts therefore remain two-valued, but the consequence relation is non-classical: excluded middle fails, and double negation elimination does not hold. The system computes consequences only where the relevant reading forces them. These are machine-checked classifications, not interpretations. The broader warrant-containment work then represents claims and their dependencies so that the removal or expiry of a ground can affect what rests upon it.

For legal systems, the value of this approach is not that it solves legal interpretation. It makes explicit a discipline that legal AI frequently violates: a conclusion should not inherit more warrant than its grounds can support. If an output rests on a precedent, factual certification, delegated authority, or policy state, the status of that ground must remain connected to the output.

3.2 Prospective propagation

The important temporal shift is from forensic reconstruction to prospective containment. A conventional audit can discover after the event that an agent relied on an expired authorization or a displaced precedent. A dependency-aware system can instead identify what is put at risk when the change is registered. It can name the conclusions that require revalidation before they are used again.

This is especially relevant to legal memory because the cost of staleness is not adequately measured by elapsed time alone. A proposition can remain stale for months without consequence if nobody relies on it. A short period of staleness can be severe if it supports many consequential decisions. Reznik's experiments frame exposure through actions taken while containment conditions are not satisfied. In the reported synthetic model, delay alone was sufficient to produce a high share of actions outside the system's own containment requirement. The number is not transferable as an estimate of legal systems, but the structural lesson is: decision frequency and dependency reach matter more than the age of a stale record in isolation. The experiment separating delay from consequential actions was designed by Arkadiy Miteiko and implemented in Reznik's repository.

3.3 Brackets instead of false precision

Another useful discipline concerns uncertain independence. Two grounds may appear independent because they are represented as separate documents, while both derive from a common source. A legal opinion and a compliance report may repeat the same unverified factual assertion. Two judicial opinions may copy a mistaken citation from the same secondary authority. Counting them as separate support creates false confidence.

Where independence cannot be established, Reznik's approach preserves a bracket rather than returning a single number. The lower reading treats the grounds as dependent; the upper reading treats them as independent. The width of the interval represents the price of the unresolved assumption. This has a direct legal analogue: uncertainty about evidentiary independence should remain visible in the reliance decision instead of being smoothed into a confidence score.

3.4 Relation to truth maintenance, belief revision, and provenance

Warrant containment does not begin from an empty intellectual field. Truth maintenance systems have long represented beliefs together with the reasons that support them, allowing a program to revise dependent beliefs when an assumption changes (Doyle 1979). The AGM account of theory change formalized contraction and revision of belief sets under rationality constraints (Alchourrón, Gärdenfors, and Makinson 1985). Database provenance research later developed formal ways to represent how outputs derive from inputs, including semiring approaches that preserve alternative and conjunctive derivations (Green, Karvounarakis, and Tannen 2007; Cheney, Chiticariu, and Tan 2009).

These traditions matter for two reasons. First, they prevent exaggerated novelty claims. Separating a proposition from its grounds, retaining alternative derivations, and revising consequences after withdrawal are not new ideas in themselves. Reznik expressly revised his deposited note after testing some proposed distinctions against ProvSQL and the earlier provenance literature. The present paper accordingly makes no claim to have invented dependency-aware reasoning.

Second, the legal problem is not exhausted by those formal traditions. A belief revision system can contract a proposition supplied to it. A provenance system can describe how a result was derived. Neither alone establishes the institutional criteria that make a statute authoritative, a precedent applicable, evidence sufficient, or reliance professionally justified. Those criteria are partly external to the calculus and may themselves be contested.

The distinctive contribution proposed here lies in the interface. A legal normative-memory architecture should use formal containment to preserve and propagate represented dependencies while reserving authority admission, admissibility, and reliance for explicit institutional decisions. This is not a claim that law is beyond computation. It is a claim that the inputs and outputs of legal computation must be governed by a structure capable of representing who is entitled to decide, under which legal order, for what purpose, and with what consequence.

4. The Incomplete-Map Problem

4.1 Incompleteness does not have one safe direction

The most important correction in Reznik's deposited note concerns incomplete dependency maps. It may be tempting to assume that missing edges always make a system overconfident. That is incorrect.

A missing shared origin can make two grounds appear more independent than they are. The measured blast radius is then too small because the system fails to see that both grounds can fall together. This is an optimistic error. It may allow action on a warrant the institution does not truly hold.

A missing alternative ground has the opposite effect. The system may believe that a conclusion has lost support even though another valid ground remains. The measured blast radius is then too large. This is a pessimistic error. It may cause unnecessary refusal, delay, or institutional idleness.

The graph itself cannot determine which kind of omission it contains. The direction of error therefore cannot be assumed. This result has significant consequences for legal AI. Completeness cannot be represented by a generic confidence label. A system must disclose what kinds of dependencies it was designed to capture, what sources were available, and where omissions could produce either unsafe reliance or excessive refusal.

4.2 Legal relevance is partly constituted outside the graph

Dependency mapping is not merely an extraction problem. Legal dependencies are often interpretive. A later decision may rely on an earlier principle without citing it. A contractual conclusion may depend on a mandatory norm that no party mentioned. A factual premise may become legally decisive only after a particular characterization is adopted. The relevant graph changes as legal hypotheses change.

This means that no fixed dependency map can be presumed complete. Legal reasoning often follows an iterative cycle: provisional characterization, identification of missing facts, retrieval of authority, revision of the characterization, and renewed testing. Normative memory must support this iteration and record which dependencies were asserted, inferred, contested, or left unresolved.

4.3 The honest output may be indeterminate

When the system lacks the information needed to decide whether a ground is independent, current, or authoritative, the correct output is not a default approval. Nor is it necessarily a categorical rejection. The appropriate state may be indeterminate, accompanied by the specific missing condition and the next inquiry capable of resolving it.

This is a governance function, not merely a stylistic preference. Material uncertainty must alter workflow state. A legal AI system that labels an output "low confidence" but allows it to enter the same downstream process has not governed uncertainty. It has described it. Living normative memory should instead route the affected conclusion to constraint, escalation, revalidation, or refusal.

5. A Five-Layer Architecture for Living Normative Memory

The relationship between formal containment and legal governance can be represented through five distinct layers.

5.1 Layer 1: Authority admission

The first layer decides what may enter the system as a legally relevant ground. It records source identity, jurisdiction, issuing body, temporal validity, hierarchical status, applicability conditions, and the actor responsible for its admission. This is where a statute, precedent, administrative act, contractual clause, evidentiary record, or delegated authorization is recognized as a candidate ground.

The admission decision is not permanent. A source can later expire, be revoked, displaced, distinguished, amended, or rendered inapplicable. The system must therefore preserve admission as a versioned institutional act, not as an intrinsic property of the document.

5.2 Layer 2: Dependency representation

The second layer represents how interpretive conclusions depend on admitted grounds. Dependencies may include conjunctive grounds, alternative grounds, exceptions, defeaters, factual predicates, jurisdictional assumptions, and professional judgments. Each edge should identify its status and provenance. Some dependencies will be explicit and verified. Others will be proposed, inferred, contested, or unknown.

This layer should also preserve claims of independence as claims, rather than facts granted by default. If two grounds are treated as independent, the basis for doing so should be inspectable. Where independence cannot be verified, the system should preserve the resulting bracket.

5.3 Layer 3: Formal containment

The third layer computes the effects of a registered change across the represented dependency structure. If a ground is withdrawn, the system can identify conclusions that lose support, conclusions that survive through alternative grounds, and decisions that require renewed evaluation.

This is the natural domain of ZTL and related provenance or truth-maintenance approaches. Its output is conditional: given the graph and the change supplied, these nodes are affected. It does not certify that the graph is complete or that the supplied legal change is correct.

5.4 Layer 4: Runtime admissibility

The fourth layer asks whether an affected interpretation remains fit for use under the conditions existing at the moment of reliance. It considers current authority, facts, jurisdiction, policy state, evidentiary sufficiency, review requirements, and the consequence contemplated.

Runtime admissibility is not reducible to formal derivability. A conclusion may still follow from recorded premises while being inadmissible because the actor lacks authority, the matter falls outside the jurisdiction, the evidence is insufficient for the intended consequence, or professional review is required. Conversely, the withdrawal of one ground may not defeat reliance if an independent legal basis remains and has been properly reviewed.

5.5 Layer 5: Institutional reliance

The fifth layer records who may rely on the conclusion, for what purpose, under what responsibility, and with what continuing effect. A legal output may be admissible for internal exploration but not for client advice, regulatory filing, adjudication, payment, denial of a claim, or another binding act.

Reliance is therefore actor-specific and consequence-specific. The same interpretation can occupy different states for a researcher, an in-house legal team, a regulator, a court, or an automated agent. The system should preserve the decision that converted an interpretation into an institutional position, together with the authority and review that supported that conversion.

6. Legal Scenarios

6.1 Overruled precedent with surviving doctrinal descendants

Assume that Decision A establishes an interpretation subsequently adopted in Decisions B, C, and D. B cites A directly. C cites B. D contains the same rule but cites only a secondary treatise. Decision E later overrules A.

A citation-only system can readily flag B and perhaps C. D is harder because its dependency is phenotypic rather than bibliographically explicit. A formal containment system can propagate through the edges it possesses, but the unrecorded relation to D remains invisible. CriptoIUS should therefore distinguish direct citation, asserted doctrinal inheritance, semantic similarity, and professionally validated dependence. Semantic similarity may propose an edge. It cannot silently create one with legal effect.

The revalidation result may also differ among descendants. B may lose its sole ground and be blocked for future reliance. C may survive because it added an independent statutory basis. D may require escalation because the system can detect a suspiciously similar rule but cannot establish the lineage. The overruling event does not erase any decision. It changes their prospective reliance state while preserving their historical role.

6.2 Statutory amendment with transitional law

Assume that a statute is amended on 1 January. The new text governs contracts formed after that date, while earlier contracts remain subject to the previous regime. A conclusion based on the former text is not simply true before the amendment and false afterward. Its warrant becomes temporally partitioned.

A living normative-memory system must represent the amendment, effective date, transitional provision, relevant event date, and characterization of the legal relationship. The containment event identifies outputs that relied on the old text. Runtime admissibility then determines which remain usable for earlier contracts and which must be revalidated for later ones.

This example shows why expiry cannot be treated as deletion. The prior interpretation remains legally operative within a bounded temporal domain. A system that globally replaces it with the amended rule would produce the same kind of error as a system that never updates at all.

6.3 Jurisdictional migration

Consider a contract template developed under New York law and reused in an Argentine transaction. The clause remains textually intact, but its legal meaning can change when mandatory Argentine law, local interpretive practices, and procedural rules become relevant. No source needs to be repealed for the original warrant to fail. The change occurs in the relation between source, actor, transaction, and governing legal order.

In graph terms, the relevant authority and applicability predicates have changed. In legal terms, the interpretation must pass through the receiving normative system. The original source can remain persuasive or commercially useful without retaining the authority it possessed in its home jurisdiction.

CriptoIUS should record jurisdictional divergence as a first-class event and create a locally reviewed child interpretation rather than allowing the source template to masquerade as universal law. This is also where EPT becomes explanatory: successful templates replicate across borders because they reduce transaction costs, even when their inherited legal assumptions do not fit the receiving environment.

6.4 Expired evidence and continuing decisions

Assume that an institution approves a counterparty on the basis of a certificate valid for one year. The approval supports later payments, delegations, and automated permissions. When the certificate expires, the relevant exposure is not the number of days since expiration. It is the set of consequential acts that continue to rely on the approval.

A dependency-aware system can identify those acts if the certificate, approval, and permissions are connected. Runtime admissibility must still decide whether expiry automatically blocks action, permits a grace period, or requires alternative evidence. The graph calculates reach. The governing policy and law determine consequence.

6.5 Withdrawal of one ground among several

Assume that a legal conclusion rests on both a statutory interpretation and an independent contractual waiver. If the statutory ground is displaced, the conclusion may survive through the waiver. If the waiver is invalid under mandatory law, however, the apparent alternative is not legally independent.

This scenario reveals why formal alternatives and legal alternatives are not identical. A graph may show two incoming grounds. Legal review must establish whether each is independently capable of supporting the consequence. Until that is resolved, the proper representation is a bracket or an escalation state, not an unqualified declaration of redundancy.

6.6 Conflicting institutional reliance

The same interpretation may be adequate for internal research, insufficient for client advice, and prohibited as the sole basis for an administrative sanction. The underlying proposition has not changed. The consequence and the institutional standard have.

Normative memory should therefore avoid one universal validity flag. It should record reliance decisions by actor, purpose, and consequence. This prevents a low-stakes exploratory output from acquiring authority merely because it is reused downstream in a more consequential setting.

7. Public Design Principles

The preceding analysis supports several technology-neutral principles. A consequential legal system should preserve the historical basis of an interpretation, make material changes in that basis visible, distinguish past validity from present fitness for reliance, and require an accountable legal decision where formal propagation cannot resolve authority or consequence.

These principles do not determine a product architecture. They remain compatible with different storage models, verification techniques, review arrangements, and institutional settings. The concrete implementation within CriptoIUS, including data structures, workflow states, thresholds, scoring, access controls, and product sequencing, falls outside the scope of this public paper.

7.1 The role of human legal judgment

Human review is not added because formal systems are inherently untrustworthy. It is required because the relevant legal predicates are institutionally constituted. Whether a decision overrules another, whether a statutory amendment applies retroactively, whether two sources are independent, or whether evidence is sufficient for a particular legal consequence are legal judgments. They can be assisted by computation, constrained by formal rules, and documented in structured form. They cannot be created solely by graph propagation.

The responsible reviewer must therefore be able to inspect and overturn the system's dependency assumptions. Otherwise authority migrates silently from legal reasoning to the architecture that selected and connected the grounds.

8. Failure Modes and Safeguards

8.1 Hidden common origin

Two apparently independent grounds may derive from the same report, dataset, model output, or mistaken citation. The safeguard is not to assume independence from documentary plurality. The record should preserve source lineage where known and represent unresolved independence through a bracket.

8.2 Missing alternative ground

A system may block a valid interpretation because it failed to record an alternative source of authority. The safeguard is a revalidation workflow that permits retrieval and professional supplementation before final refusal.

8.3 Stale authority state

A dependency graph can be internally coherent while its authority metadata is outdated. The safeguard is event-driven and periodic review of high-consequence grounds, with priority determined by downstream reliance exposure rather than age alone.

8.4 Self-reported compliance

An agent may generate its own execution record and declare that its action was authorized. Authentication proves origin, not truth. The safeguard is independent verification of the authority and evidence state, together with a record of the human or institutional actor who accepted reliance.

8.5 False certainty from scalar scores

Confidence, reliability, or blast-radius scores can conceal assumptions about independence and completeness. The safeguard is to expose ranges, unresolved assumptions, and the concrete conditions that would change the result.

8.6 Governance gates that merely work often

Reznik reports that the tested gate held violations at zero on three channels and leaked on two. The margin was therefore not a sound bound. This is an important warning for legal AI. A control that usually prevents an inadmissible consequence may still be unacceptable where the consequence is irreversible or legally significant. The appropriate response is not to hide the leakage behind aggregate accuracy, but to identify the regimes in which the gate fails and route those cases to a different workflow.

9. Evaluation Protocol

A future implementation should be evaluated against adversarial legal scenarios rather than only clean dependency graphs. At minimum, the test set should include:

1. a precedent later overruled, with downstream interpretations that cite it directly;
2. a hidden shared source reproduced across multiple opinions;
3. an unrecorded alternative statutory ground that preserves the conclusion;
4. a jurisdictional transfer that changes applicable authority;
5. a factual certification that expires while dependent decisions continue;
6. a legal amendment with transitional provisions preserving earlier cases;
7. a malicious or erroneous input that falsely marks authority as current;
8. a conclusion admissible for internal research but not for external reliance;
9. a dependency discovered only after a provisional legal characterization; and
10. two systems producing identical conclusions from materially different warrant structures.

Evaluation should measure more than final-answer accuracy. Relevant metrics include dependency recall, false independence, missed alternative grounds, time to containment, consequential actions taken while stale, appropriate abstention, quality of the revalidation request, and the ability of a reviewer to reconstruct and overturn the system's decision.

No single benchmark can establish institutional adequacy. The benchmark itself must disclose which dependencies are complete by construction, which authority changes are supplied as ground truth, and which legal judgments are performed by human annotators. Otherwise the evaluation hides the very legal work the system claims to automate.

10. Evolutionary Hypotheses and Research Agenda

The integration of EPT, normative memory, and warrant containment produces a set of hypotheses that can be tested rather than treated as metaphorical claims.

10.1 Persistence after formal withdrawal

Hypothesis 1: Interpretive patterns embedded in templates, software, and professional routines will continue to appear in downstream legal work after their originating authority is formally displaced.

This can be tested by identifying a population of overruled or amended legal rules and measuring their continued presence in later contracts, opinions, administrative forms, and machine-generated outputs. The relevant outcome is not mere textual similarity. Researchers should distinguish direct citation, copied formulation, functional equivalence, and independently reconstructed reasoning.

10.2 Infrastructure increases phenotypic persistence

Hypothesis 2: Interpretations encoded in reusable infrastructure will decay more slowly after authority withdrawal than interpretations preserved only in ordinary legal texts.

The proposed mechanism is switching cost. Once an interpretation is embedded in a workflow, schema, checklist, or automated decision rule, removing it requires more than updating a source database. The institutional phenotype can therefore outlive the legal warrant that originally produced it.

10.3 Dependency visibility changes revalidation speed

Hypothesis 3: Systems that preserve validated dependency edges will identify affected downstream uses faster and with fewer omissions than systems relying only on semantic retrieval or citation search.

The comparison should include both optimistic and pessimistic error. A system that flags everything may achieve high recall while imposing excessive institutional idleness. A meaningful evaluation must measure unsafe continued reliance, unnecessary withdrawal, and time to responsible revalidation.

10.4 High replication is not a proxy for authority

Hypothesis 4: Highly replicated interpretations will sometimes retain high usage after losing formal authority, particularly where users face weak review incentives or cannot observe the lineage of the rule.

This hypothesis directly separates evolutionary fitness from legal validity. If confirmed, it would show why adoption metrics such as JurisRank must remain subordinate to an authority and admissibility gate. Popularity can reveal institutional fitness while concealing normative obsolescence.

10.5 The cost of staleness tracks consequences, not duration

Reznik's repository already probes the technical distinction between elapsed steps and consequential acts during a stale interval, using an experimental design proposed by Arkadiy Miteiko. The open legal hypothesis is therefore narrower:

Hypothesis 5: In real institutional settings, harm from stale legal warrants will correlate more strongly with the number and type of dependent decisions than with the elapsed period of staleness alone.

Testing requires event-level records linking changes in authority to later acts of reliance. A short-lived defect in a high-volume automated workflow may matter more than a long-lived defect in an unused archive.

10.6 Research design

A credible research program should combine doctrinal coding, graph analysis, controlled synthetic cases, and retrospective institutional records. It should pre-register the legal criteria for authority change, distinguish observed dependencies from model-proposed ones, and preserve human disagreement rather than forcing a false gold standard.

The program should also test transferability across legal systems. Rules of precedent, statutory temporal effect, administrative validity, and evidentiary sufficiency differ among jurisdictions. The proposed five-layer architecture is intended to be portable, but its authority predicates must be locally specified and independently reviewed.

11. Objections

11.1 Legal reasoning cannot be reduced to a graph

This objection is correct if the proposal is understood as a reduction. Legal reasoning includes interpretation, analogy, characterization, institutional practice, contested values, and open-textured standards. The proposed graph does not replace those operations. It records the dependencies that an institution has chosen to make explicit and computes consequences conditional on that representation.

The value of the graph is therefore procedural and epistemic. It makes some assumptions inspectable, supports prospective revalidation, and exposes what the institution has failed to represent. Its incompleteness must remain an express limitation, not a defect hidden behind mathematical notation.

11.2 Human review merely relocates the bottleneck

Human review can become ceremonial, inconsistent, or too costly to scale. The answer is not to remove it from decisions whose legal predicates require institutional judgment. It is to allocate review according to consequence, uncertainty, and dependency reach.

Formal containment can reduce the review burden by identifying what changed and which outputs are affected. Structured revalidation can require the reviewer to address authority, jurisdiction, evidence, alternatives, and permitted reliance. The goal is not a human signature on every computation. It is accountable judgment where computation cannot supply the governing premise.

11.3 Append-only records may preserve error

They do, and that is partly the point. An institution must be able to show what it believed and why, even when the belief was later corrected. Append-only architecture should preserve the error together with the event that displaced it, not continue presenting the error as current.

The alternative, silent overwriting, damages both historical reconstruction and accountability. Versioning does not legitimate the old interpretation. It separates historical existence from present admissibility.

11.4 The model may be too costly for ordinary legal practice

Full dependency representation would be disproportionate for many low-risk tasks. The architecture should therefore scale with consequence. High-impact, repeatable, or automated interpretations justify richer dependency and revalidation records. Exploratory research may require only source provenance, jurisdiction, temporal status, and an explicit prohibition on unreviewed downstream reliance.

This proportionality principle also prevents governance from becoming process theater. The objective is not maximal documentation. It is sufficient evidence to justify reliance and contain material changes.

11.5 Institutions can manipulate the authority inputs

Yes. Reznik's instrument is inert against an adversary that controls the input and continues to sign false statements. A legal governance system cannot solve institutional bad faith through dependency propagation alone.

The architecture should therefore separate the actor proposing an authority state from the actor permitted to validate it, preserve contradictory evidence, and make material overrides visible. These controls reduce opportunity for undetected manipulation. They do not abolish fraud, capture, or abuse of power.

12. Discussion

12.1 Formal containment and legal admissibility are complementary

The central boundary can now be stated precisely. Formal containment answers: if this represented ground changes, what represented conclusions are affected? Runtime admissibility answers: given the actual legal and institutional conditions now in force, may this conclusion still support the contemplated reliance?

Neither question replaces the other. Without containment, legal governance is forced to reconstruct dependency cascades manually and often too late. Without admissibility, formal propagation can produce technically coherent decisions that lack institutional validity.

12.2 Normative memory is not a truth machine

The proposed architecture does not promise automated legal truth. It promises a more disciplined account of why an interpretation was available for reliance, what it depended on, and what happened when those conditions changed. That is a narrower claim and a more useful one.

Legal disagreement will remain. Different institutions may admit different authorities, characterize dependencies differently, or reach different revalidation decisions. CriptoIUS should preserve those divergences rather than force premature consensus. The infrastructure becomes valuable by making the grounds and transitions inspectable, not by erasing normative pluralism.

12.3 Evolution requires inheritance with defeasibility

CriptoIUS treats legal interpretations as evolving institutional artifacts. Evolutionary persistence, however, must not be confused with continuing legal authority. A widely replicated interpretation can become obsolete, unlawful, or inapplicable. Fitness metrics may describe adoption and dispute patterns, but they cannot independently establish legitimacy.

Warrant containment adds a missing discipline to evolutionary legal memory. Interpretations inherit structure, but their entitlement to guide future action remains defeasible. When a ground changes, the lineage does not vanish. It becomes the object of revalidation.

12.4 The architecture changes professional responsibility

A lawyer relying on an AI-supported conclusion should not be asked merely whether the source existed or the model was accurate. The relevant record should show which version entered the matter, what authorities and facts supported it, whether material dependencies remained current, what uncertainty was unresolved, and who authorized its use.

This is a record of reliance, not just an audit log. It is generated when the interpretation becomes an institutional position, not assembled later from technical traces designed for debugging. The distinction aligns system design with the professional question that courts, clients, boards, and regulators are likely to ask: why was this output fit to rely on at that time and for that consequence?

13. Limitations

This paper is conceptual. It proposes an architecture and an evaluation protocol; it does not report a completed CriptoIUS implementation of ZTL or a validated deployment in legal practice.

The technical evidence is also bounded. Reznik's containment measurements are simulations on synthetic graphs. The deposited note expressly reports that the tested real graph differed materially in density, structure, and redundancy. The existence of a structural effect may be informative while its measured location and magnitude remain model-dependent.

The proposed architecture cannot eliminate incomplete legal knowledge. It can make recorded dependencies inspectable and omissions consequential, but it cannot discover every hidden source, legal assumption, or factual relation. Nor can it determine legal authority without institutionally supplied criteria and responsible human judgment.

Finally, append-only records and cryptographic proofs can support integrity and reconstruction, but they do not establish legal correctness. Confidentiality, data protection, procedural fairness, and access control require separate design and legal analysis.

14. Conclusion

Legal AI systems are beginning to preserve more than documents. They preserve embeddings, citations, agent traces, and execution records. The next step is not simply more memory. It is governed normative memory.

A living normative memory system must know which conclusions depend on which grounds, preserve the institutional act that admitted those grounds, and react when their status changes. Formal warrant containment can compute the prospective effects of a represented withdrawal. It can expose blast radius, alternative support, staleness, and unresolved independence. It cannot decide whether a legal withdrawal truly occurred, whether the graph is complete, or whether an affected conclusion may still bind an institution.

Those decisions belong to authority admission, runtime admissibility, and institutional reliance. CriptoIUS can provide the versioned interpretive infrastructure that connects them without collapsing their differences.

The governing principle is therefore not that every legal conclusion must be permanently certain. It is that no conclusion should continue to exercise institutional force after its warrant has changed without making that change visible, tracing its dependencies, and requiring a renewed decision about reliance.

Acknowledgements

The author thanks Vitaly Reznik for his generous and intellectually rigorous exchange, for correcting and delimiting the technical claims discussed in this paper, and for his willingness to read this working paper. His insistence on publishing negative results and express limitations materially improved the analysis. The distinction developed here between formal warrant containment and legal admissibility emerged through an independent exchange between the authors. Any legal synthesis, application to normative memory or CriptoIUS, or remaining error is solely the responsibility of the present author.

The author also thanks Arkadiy Miteiko for introducing Reznik's work and for designing the experiment, implemented in Reznik's repository, that separates elapsed delay from consequential actions taken while a dependency map remains stale.

References

Alchourrón, Carlos E., Peter Gärdenfors, and David Makinson. 1985. "On the Logic of Theory Change: Partial Meet Contraction and Revision Functions." *Journal of Symbolic Logic* 50 (2): 510-530.

Cheney, James, Laura Chiticariu, and Wang-Chiew Tan. 2009. "Provenance in Databases: Why, How, and Where." *Foundations and Trends in Databases* 1 (4): 379-474.

Doyle, Jon. 1979. "A Truth Maintenance System." *Artificial Intelligence* 12 (3): 231-272.

Green, Todd J., Grigoris Karvounarakis, and Val Tannen. 2007. "Provenance Semirings." In *Proceedings of the Twenty-Sixth ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*, 31-40.

Lerer, Ignacio Adrián. 2025. *CriptoIus: A Global Evolutionary Legal System Through Blockchain: How Extended Phenotype Theory and Compatibilist Philosophy Enable Self-Organizing Legal Order Beyond Contracts*. Working paper. Public architecture repository: https://github.com/adrianlerer/Criptoius-Open.

Reznik, Vitaly. 2026. "Warrant Containment in Dependency Graphs." Current version. Zenodo. https://doi.org/10.5281/zenodo.21981726. See especially sections 3.6 and 6.